larpix-lib on PACMAN

LArPix control software: architecture, setup, bench results and status

Maintained by the PACMAN agents (augie)

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1 Purpose and scope

`larpix-lib` (package `larpix_control`) is the new LArPix control software for PACMAN with the zipline firmware and software (`zipline.tex`). Since 2026-11-05 it is a tool Michael supports officially, no longer an experiment, and this record moved back into pacman_docs `agents/` beside `zipline.tex`. The goal is for `larpix-lib` to replace (1) the control path, including readback for control, and (2) the data server; the focus is (1). The ASICs themselves (bench setup, registers, TX settings) are recorded in `larpix.tex`.

Scope (Michael 2026-11-05): a single Hydra strand on one UART channel, taken far; more than one channel is out of scope for now (the binding of io groups and channels to strands is postponed), but the design must scale to the whole system. Every PACMAN carries precisely one ASIC type, so the ASIC version is an attribute of the PACMAN.

2 Repository and environment

- `mulhearn/larpix-lib`, cloned in `/agents/cowork/larpix-lib` (shared repository; Michael pushes). Work is on the branch `working` (no releases yet; `main` follows `origin/main`). At 3f0bbaa: PEP 8 names (e1e2dbb), David Yang's correlator (fea7566), the closed loop (322c119), board control and `demos/hello_asic.py` (2c3e19f), the register dump and 16000-word requests (4f84da5), David Yang's data packet layout (da768fc), rails, the data spigot, pedestals and the occupancy view (3f0bbaa). Needs PACMAN software 3.6.3 (requests over 100 words per UART).
- Python environment: conda environment `larpix` per user, an editable install of the clone; `make test` runs the unit tests (46, OK).
- Names (Michael 2026-11-05): Python classes in CapWords (PEP 8), functions, methods, modules snake_case, constants UPPER_CASE. "PACMAN" names only the real board or an accurate emulation; toys and generic contracts say "board".

3 Architecture

`larpix_control` separates what varies independently: the ASIC version, what each configuration step means, the network topology, register access to the ASICs, and control of the board they sit on. Version-specific content lives in YAML, not code. The contracts are abstract base classes in `common.interfaces`.

Layer	Module	Role
ASIC spec	`asic.asic_spec`	One ASIC version from YAML (`config/asics/larpix_v3.yaml`): register space, config and data packet layouts. Field updates to register writes (`build_write_list`), registers to config packets, the read side (`valid_config_read_response`, `parse_chip_address_value`) and data packets (`valid_data_packet`, `parse_data_packet_fields`). No I/O.
Fragments	`asic.fragment_lib`	Named configuration steps in YAML (`config/fragments/`) with external parameters and per-version overrides.
ASIC config	`AsicConfigIface`, `asic.asic_config`	The operations a network needs (chip ID, I/O init, input and output enables), and any named fragment (`apply_fragment`), one step up from register access. `AsicConfig` evaluates a fragment, turns the field updates into `RegisterOp` writes on its io channel and transmits them.
Topology	`hydra.strand`	One Hydra network on one UART, from YAML (`config/hydra/`); configures it from the root outward. Uses only `AsicConfigIface`.
Register access	`AsicRegisterIface`	`transmit(ops, fire_and_forget)`: `RegisterOps` (`OpType` WRITE or READ, `nrep` repeats, io channel, chip, register, value) in order; returns `OpReplies`: per register (`RegisterKey`) the `WriteReply` (requested, acknowledged) and `ReadReply` (requested, acknowledged, majority value, received, discrepant), and an overall `complete`.
Board control	`BoardControlIface`	One readout board (one io group): `ping`, `get_tile`, `reset` (FULL or INTERNAL, by tile) with `wait_on_reset`, `power_on`/`power_off` (by tile), `rail_off`/`rail_on` (VDDA or VDDD, by tile), `uart_enable`/`uart_disable` (by io channel). Pokes return at once, without replies.
Data	`AsicDataIface`	Everything the ASICs on one board send, as it arrives: `start_data`, `read_data(timeout_ms)` (a list of `RxWord`: io channel, 64-bit word, the board's receive time stamp), `stop_data`; read regularly (a stalled subscriber makes `pacman_dataserver` quit).
PACMAN	`pacman.pacman`	`Pacman` implements all three: one PACMAN from one network file entry; ASIC words and board register writes through `pacman_cmdserver` (each acknowledged), replies from `pacman_dataserver` matched by the reply correlator (`asic._correlator`). `load_network` builds one per entry; `sync_pacman_id` sets the PACMAN's own ID to its io group; up to 16000 words per request; the data spigot on a SUB socket of its own; rails by DAC (0 or the `pacman_menu` nominal).
Toy	`common.board_toy`	`BoardToy` implements all three without hardware: registers in memory, resets, power, rails and UART enables with their effect on the chips; no data (simulation is for David Yang's simulator).

A configuration runs: `Strand.init_network` → `AsicConfig` operation → fragment → field updates → register writes → `RegisterOps` → `Pacman.transmit` → config packets → `pacman_cmdserver`; replies return through `pacman_dataserver` to the correlator and the `OpReplies`.

3.1 Design decisions (Michael 2026-11-05)

- Chip identity is `(io_chan, chip_id)`: chip IDs are unique only within an io channel. Each board owns a configured range of io channels (`first_io_chan`, `uart_mask`); the upper bits may encode the PACMAN by configuration, never checked. The wire channel is the io channel's low byte (UART + 1); the UART is `io_chan - first_io_chan`, so no "+1" appears in our code.
- One board per io group: the board knows its io group, so board control takes tiles and io channels

only.

- Replies: one persistent SUB socket per PACMAN, subscribed only during a transmission (an idle subscriber cannot stall the data server) and drained at its start (no stale reply counts as fresh). The race between the subscription and the first reply is accepted until the 3.7 XPUB confirmation.
- The timeout is a property of each implementation (`timeout_ms` in the network file, 200 ms), bounding the acknowledgements and the wait for replies. Missing acknowledgements and replies are reported, not raised. `fire_and_forget` skips collecting replies; acknowledgements are still counted.
- Reads repeated (`nrep` or listed again) are collated: the majority value (the last received on a tie) and the number discrepant.
- PACMAN registers, power values and the reset pokes are private to `Pacman`, from `pacman_vspace.cc`. The code there sends the internal reset at 0x00100410 and the full reset at 0x00100420; its header comment has the two swapped.
- Interface first, in Python: the interface step reviews generated `.pyi` stubs (baseline committed on the interface branch, the draft in the stubs, propagated into the `.py` once accepted); the stubs never reach `working`.

Later decisions (Michael 2026-11-05, evening).

- Interfaces are reviewed in Python as generated `.pyi` stubs (`pyi-baseline.py`): a baseline of the modules a topic changes committed on the interface branch, the draft in the stubs, then propagated into the `.py`; configuration schemas (YAML) reviewed with them.
- The data spigot is its own ABC, not board control; the ASIC configuration for a task is a fragment applied with `apply_fragment`.
- Data packet fields that diagnostics overlay on the timestamp carry `enabled_by` (the FIFO counts, the first-packet marker) or `disabled_by` (the timestamp); parsing defaults to nothing enabled, and this version enables none.
- Error replies of `pacman_cmdserver`: one code, 0xEEEE, the cause in its log.
- The HDF5 data server is a separate component (nothing in `larpix_control` depends on it), tabled; the occupancy monitor belongs to `larpix-lib`.

4 Network file

`config/network/*.yaml`, one entry per PACMAN:

Field	Meaning
`io_group`	The PACMAN; set as its own ID by `sync_pacman_id`.
`hostname`, `port`	`pacman_cmdserver` (5555).
`data_port`	`pacman_dataserver`; optional, default 5556.
`timeout_ms`	Acknowledgement and reply timeout (200).
`first_io_chan`, `uart_mask`	The board's io channels: `first_io_chan` + u for each bit u; the low byte of each must be its PACMAN channel.
`uarts_per_tile`	4 on every revision (1v5: 10 tiles, 40 UARTs; 1v4: 8 tiles, 32 UARTs).

The bench file is `single_bench.yaml`: the 1v5 at 10.42.0.175, io group 1, `first_io_chan` 0x001, `uart_mask` 0xF (tile 0, io channels 1–4).

5 Hello ASIC

`demos/hello_asic.py`, from the repo top, takes a PACMAN just booted or powered up to a configured, answering strand:

1. Ping until `pacman_cmdserver` answers (up to `-wait`, 120 s), then `sync_pacman_id`.
2. On the strand's tile: UART off, `power_on`, 50 ms, full reset, `wait_on_reset` (1 ms).
3. Configure the strand (`Strand` → `AsicConfig` → `Pacman`), recording every register written; UART on.
4. Read back every register written and each chip's ID register; PASS or FAIL, and the exit status.

Options: `-network`, `-io_group`, `-io_chan`, `-hydra`, `-asic`, `-wait`, `-v` (the configuration trace and the board's warnings). No RX buffer enables are written: the correlator ignores everything but config replies. It replaces the earlier `agents/bringup.sh` and `pacman_util.py`.

The settles come from a sweep on the 1v5 (tile 0 off 1 s before each run, 10 runs per step): a full reset passes with no settle at all (the reset is about 13 µs, shorter than the next round trip), and the power settle passes from 10 ms and fails at 5 ms and below. 50 ms and 1 ms keep margins of 5× and about 75×; the 1v4 and other tiles are not measured.

6 Pedestals and occupancy

Pedestals copy larpix-control's `pedestal_qc.py` (larpix-10x10-scripts), in two fragments: `pedestal_trigger` (periodic and rolling periodic trigger, periodic reset, no hit veto, 100000 trigger cycles, 4096 reset cycles, register 128 complete, all 64 CSAs on) and `pedestal_channels` (trigger and channel masks cleared); `pedestal_stop` turns the trigger off again. The cycle counts are fixed bytes in the fragment (fragments cannot split a number across a byte array). `demos/pedestal.py` writes both to every chip of the strand with the tile's VDDA off (as larpix-control), counts the data and stops the trigger (`-keep` leaves it).

`demos/occupancy.py` is a live strand view of one io channel: a row per chip (hydra file order) and an "other" row, one character per channel (the digit n for at least 10^n hits/s, a dot under 1/s, blank for none), per row hits/s and mean ADC, refreshed each second over a sliding window. With the rolling trigger every channel sits at about 1.6 hits/s, so each cell shows 0.

7 Results

Date	Board	Result
2026-11-02	1v5	First configuration by `larpix-lib` (`demos/init_network.py` from felix, through `pacman_cmdserver`), with the input enables left out: 13 writes. Then a menu hello without the configuration root: the reply exact, 3 of 3. Before it, after a full reset, hello had no reply.
2026-11-02	1v5	With the input enables written (126 = 0x02, POSI1 only; 14 writes): the reply exact, 3 of 3; read back 124 = 0x00, 125 = 0x01, 126 = 0x02, 248 = 0x07. UART 0 drives POSI1 (west), as the port map has it.
2026-11-02	1v5	The full configuration sequence (fce6523, "ASIC config working at test bench"): the reply exact, 3 of 3; io channels 2–4 unanswered, channel 1 exact.
2026-11-03	1v5	The whole chain, by Michael, on the software release 94e3f01: the configuration by `larpix-lib` through `pacman_cmdserver`, the menu's hello replies seen on felix through the data server.
2026-11-04	1v5	Bring-up from felix alone (then `agents/bringup.sh`): tile 1 power (VDDA 2367 mV, VDDD 1302 mV), full reset, 14 configuration writes, read-back through the data server. One read at a time exact; four in a row, only the first and last replies: the data server's PUB socket dropped them (send high-water mark 1).
2026-11-04	1v5	`PUB_HWM` 1000: four reads in a row 4 of 4, 64 in a row 64 of 64; `pacman_push` 20000 words, all counted (6680 with HWM 1). In software 3.6.2.
2026-11-05	1v5	The closed loop (`AsicRegisterIface`, `PacmanAsicRegister` then): 4 reads answered in 6 ms (was a fixed 0.5 s subscription sleep); `nrep` 3 without discrepancy; 64 registers in one transmission in 8 ms; fire and forget acknowledged with no reply collected and nothing stale after it; pixel trim 0 written and read back, then restored; `init_network.py` through `AsicConfig`, configuration unchanged; 20 of 20 runs.
2026-11-05	1v5	Board control (`Pacman`): `read_pacman_id` 1 after `sync_pacman_id`; with UART 1 disabled a read is acknowledged with no reply (`complete` False), answered again once enabled; an internal reset of tile 0 keeps every register; a full reset silences chip 11 and `init_network.py` restores it from chip 1; tile 0 power off and on: VDDA 2365 → 29 → 2367 mV, VDDD 1302 → 8 → 1302 mV.
2026-11-05	1v5	`hello_asic.py`: PASS warm (14 of 14 registers, chip ID included); after `reboot`, PASS 29 s after the reboot (the board answered after about 22 s); after a supply power cycle (halt, `OUTP 0`, `OUTP 1`), PASS 47 s after power-on. With `timeout_ms` 200 and the swept settles: 0.18–0.19 s, warm and from a powered-off tile (6 of 6).
2026-11-05	1v5	`dump_registers.py`: 256 registers in 0.10 s, three reads each without a discrepancy; the reset values differing from the datasheet are recorded in `larpix.tex`. More than about 100 reads in one transmission lost the rest: `pacman_cmdserver` moved at most 100 TX buffers per request and the rest waited for the next request, its replies arriving late (`zipline.tex`, software 3.6.3); `WORDS_PER_MESSAGE` 100 meanwhile, 16000 once the fix was on the bench: the 256-register scan in one request, 13–15 ms.
2026-11-05	1v5	Pedestals (software 3.6.3 candidate): 100 data packets/s from chip 11, all trigger type periodic, all 64 channels evenly (the rolling trigger steps through the channels: about 1.6/s each); timestamps step by 100001 cycles (the period + 1) and fit in 28 bits; ADC mean about 257 (warm), range 248–542; the occupancy view live. The data packet layout (datasheet table 16) holds on real packets.
2026-11-05	1v5	`pacman_dataserver` quit during the evening (cause not seen: its output went to `/dev/null`); restarting it alone brought neither replies nor data back, restarting both servers (`pacman_cmdserver` first) did, the ASIC configuration and its running trigger untouched. Both servers now log to `/tmp`.

8 Hydra strands on the 5x5

Single hydra on the 5x5 (2026-11-07, Michael's scope: are larpix-lib's configurations sufficient?). The UC Davis 5x5 tile on PACMAN tile 1, the 1v5 on software 3.6.3; `demos/hydra_5x5.py`: one power-up and

full reset, each strand configured root outward, every register written read back, then 10 s of pedestals. The roots take larpix-lib's own fragments; every other chip also `init_link_5x5` (network_5x5.py's link settings, on all four ports, since `init_io` has no port). Hydra files `config/hydra/5x5_io1–4.yaml` (rows; row 5 south of 41) and `5x5_one_strand.yaml` (row 1 east from 11, each column south from it).

Step	Result
5 chips, row 11–15, io channel 1	configured; pedestals about 100 packets/s on every chip
25 chips, one strand from 11	configured in 0.3 s; 420 of 422 reads match; pedestals about 100/s on all 25
25 chips, four roots, io channels 1–4	pedestals about 100/s on all 25 (chip 13: 87); 1–2 reads missed per strand

Findings:

- `pedestal_channels` (channel mask 0) leaves every channel's own trigger on: on the pixel tile chip 11 sent 140 000 packets/s and saturated its UART. `pedestal_channels_5x5` masks them (channel mask 1, periodic trigger mask 0, global threshold 255), as `enable_pedestal_5x5.py`.
- With VDDA off (as `pedestal.py` writes the pedestal fragments) the root takes writes, the chips beyond it do not: writes are not passed from chip to chip. `hydra_5x5.py` keeps VDDA on.
- Register dumps against network_5x5.py (chips 11–13): larpix-lib leaves `i_rx_clk`, `i_rx_rst`, `i_rx_ext_trig` at 0 and `reset_threshold` at 255, the chips' power-on values (the spec's reset values: 2, 2, 2, 0); larpix-control writes every register (3, 3, 3, 0). Configuration and pedestals work either way.

Open: in the readback batch, the chip ID read of a root or its first child sometimes has no reply (once `0x04`), and the farthest chip's last read can time out; the same reads alone always answer. The data server quit three times during the session; restarting both servers brought it back each time.

9 Student work

David Yang (`dy241/larpix-lib`, 2026-06 to 08), each part as his own commit, authored with his GitHub no-reply address: his reply correlator and `AsicSpec` read-side fixes (fea7566), adapted to `AsicRegisterIface` in the next commit; his data packet layout and `valid_data_packet` (da768fc). His ASIC simulator with multi-chip grids (`sim/`) is not taken yet; it would implement the three board interfaces.

10 Open

- The HDF5 data server (tabled): separate from `larpix_control`, larpix-control's raw format suggested, the reference version open.
- The trigger period as a fragment parameter (needs a number split across a byte array).
- Why `pacman_dataserver` quit on 2026-11-05 (now logged).
- Board control: the power monitors and other replies (deferred: pokes return nothing for now); whether `power_on` alone leaves the ASICs as after a full reset (untested: `hello_asic.py` always resets).
- Deployment variations and more than one channel or board: the binding of io groups and channels to strands.
- For 3.7: the XPUB subscription confirmation, separate config and data streams, polling for reset complete.